# Supplementary Information for
# Self-Similarity in Online Networks During Social Movements


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## I. UNIPARTITE MODULARITY AND NESTEDNESS

In Figure SI.1 we show the result when calculating the nestedness and modularity coefficients for the unipartite networks. The unipartite co-occurrence networks display modular-to-nested transitions broadly consistent with those observed in the bipartite representation (Fig. 1b of the main text), although with lower resolution. For No al Tarifazo and Charlie Hebdo, a clear drop in modularity accompanied by a rise in nestedness is visible in the vicinity of the identified CTW. For 9N, however, the unipartite networks exhibit two candidate windows with comparable transition signatures, making it impossible to unambiguously identify the CTW from this representation alone. This ambiguity motivates the use of the bipartite user–hashtag networks as the primary guide for CTW identification, as discussed in Section III.A of the main text.

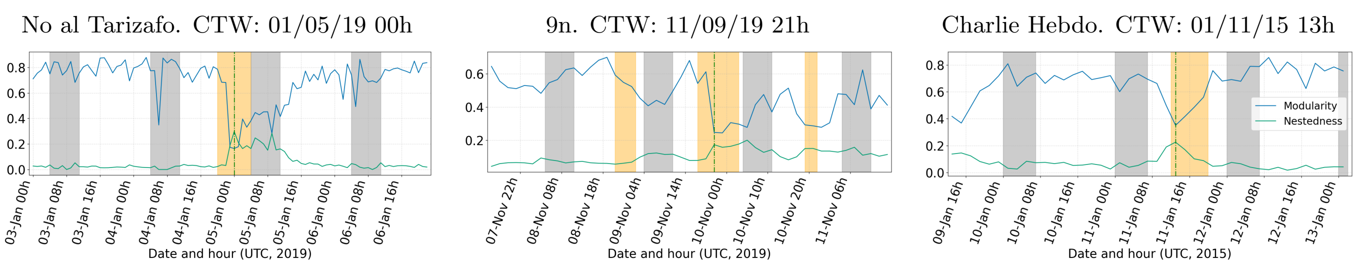


FIG. SI.1: **Nestedness and modularity coefficients for the unipartite network.** Evolution of modularity (blue) and nestedness (green) in the unipartite hashtag co-occurrence networks across temporal windows for the three social movements. Grey bands indicate low-activity periods (01:00–08:00 local time) excluded from analysis. Yellow bands highlight the identified Critical Temporal Window (CTW) for each movement, marked by a black segmented line. For No al Tarifazo and Charlie Hebdo, a modular-to-nested transition is visible at the identified CTW. For 9N, two candidate windows display comparable transition signatures, illustrating the ambiguity that motivates the use of the bipartite representation as the primary guide for CTW identification. From left to right: No al Tarifazo, 9N, and Charlie Hebdo datasets. All times are given in UTC.


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## II. $\epsilon^2$ TEST WITHOUT MOVING AVERAGE

Figure SI.2 presents the raw $\epsilon^2$ values computed for each temporal window across all three datasets, without applying the moving average smoothing procedure described in Section III-D. These unfiltered results reveal the temporal evolution of distribution collapse throughout the entire observation period for each social movement.

For the No al Tarifazo dataset (left panel), we observe considerable fluctuation in the $\epsilon^2$ values across temporal windows, with several local minima appearing throughout the time series. Importantly, whilst other minima are present in the data, many of these occur during the grey-shaded low(resting)-activity periods (01:00–08:00 local time). In contrast, the temporal window we identified as the Critical Temporal Window (CTW), marked by the vertical segmented line, is the only prominent minimum occurring during an active participation period. This distinction is crucial, indicating a qualitatively distinct state of network organisation during peak user engagement.

The 9N dataset (centre panel) also exhibits fluctuations in $\epsilon^2$ values throughout the observation period. Whilst the identified Critical Temporal Window (CTW), marked by the vertical segmented line, does not necessarily correspond to the deepest minimum in the raw $\epsilon^2$ profile, this observation is entirely consistent with the results presented in Figure 2 of the main text. As discussed in Section IV-A, the 9N movement required the convergence of multiple independent indicators—including maxima in user and hashtag participation, and the modular-to-nested transition in both co-occurrence and bipartite networks—to unambiguously identify the CTW. The $\epsilon^2$ test, both with and without moving average smoothing, represents only one component of our multi-criteria methodology.

Finally, for the Charlie Hebdo dataset (right panel), the raw $\epsilon^2$ profile also exhibits fluctuations throughout the observation period, with several minima visible across different temporal windows. However, the temporal window we identified as the CTW, marked by the vertical segmented line, represents the only significant minimum occurring outside the grey-shaded low-activity periods, consistent with the smoothed version presented in Figure 2 of the main text.

The application of the three-window moving average (N = 3), as shown in Figure 2-a of the main text, effectively filters high-frequency noise whilst preserving the fundamental temporal structure of distribution collapse. Comparison between Figures SI.2 and 2 of the main text demonstrates that the moving average procedure successfully identifies the most significant periods of self-similar organisation without introducing spurious features or obscuring genuine critical windows. The consistency between the raw and smoothed analyses provides additional confidence in our CTW identification methodology.

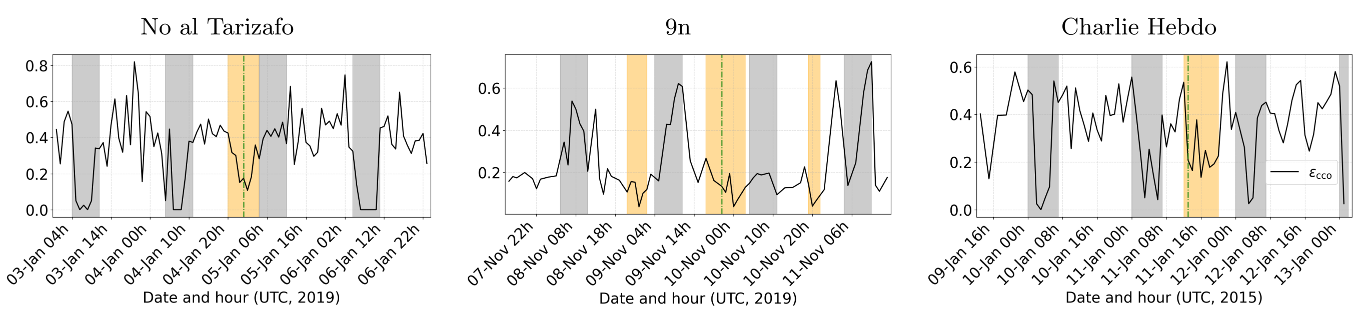


FIG. SI.2: **Raw $\epsilon^2$ values without moving average smoothing.** The $\epsilon^2$ test results for each temporal window across all three datasets, without applying the three-window moving average procedure. Grey-shaded regions indicate low-activity periods (01:00–08:00 local time), whilst yellow-highlighted segments mark candidate temporal windows for the Critical Temporal Window (CTW). The vertical segmented black line indicates the identified CTW for each movement. Despite the fluctuations in the raw data, the CTW consistently corresponds to significant minima occurring during active participation periods, validating the robustness of our multi-criteria identification methodology. From left to right: No al Tarifazo, 9N, and Charlie Hebdo datasets.

## III. COMPLEMENTARY CUMULATIVE DISTRIBUTION FUNCTION AND AVERAGE CLUSTERING COEFFICIENT FOR NOT CRITICAL WINDOWS

In Figure SI.3 we plot the same calculus shown in Figure 3 but taking time windows different from the critical one. More precisely, the selected time window for *No al Tarifazo* is Thursday, January 4, 2019 from 21:00 to 22:00 (Buenos Aires time, GMT−03:00). For the *9N* movement, Friday, November 9, 2019 from 18:00 to 20:00 (Buenos Aires time, GMT−03:00). Finally, for the *Charlie Hebdo* dataset, we set the CTW to Sunday, January 11, 2015 from 22:00 to 00:00 (Paris time, GMT+01:00)

The difference between the CTW (left column) and the non-critical windows (right column) is consistent across all three datasets. At the CTW, the curves for different degree thresholds $k_t$ collapse onto a single functional form for both the CCDF and the clustering spectrum, with $\varepsilon^2_{cco}$ values of 0.1549, 0.134, and 0.3025 for No al Tarifazo, 9N, and Charlie Hebdo, respectively. In the non-critical windows, by contrast, the curves fail to collapse and the $\varepsilon^2_{cco}$ values are substantially higher (0.4591, 0.2996, and 0.4557, respectively), reflecting the absence of scale-invariant organisation. The values quoted above are representative point measurements. The full temporal distribution of $\epsilon^2_{cco}$ across all windows is reported in Fig. SI.2 (raw) and in Fig. 2 of the main text (moving-average smoothed).

Notably, the average degree profiles shown in the insets also differ qualitatively: at the CTW the average degree rises to a well-defined maximum before declining due to finite-size effects, whereas in non-critical windows this peak is absent or poorly defined, indicating that the hierarchical filtering does not produce a coherent sequence of self-similar subgraphs. These results confirm that multiscale $k$-cut self-similarity is specific to the CTW, identified through the convergence of multiple indicators as described in Section III.A of the main text.

## IV. PERCENTAGE OF NODES AND EDGES REMAINING IN THE NETWORKS FOR EACH DATASETS AFTER FILTERING

TABLE SI.1: Percentage of nodes and edges remaining in the networks for each datasets after filtering. Their absolute numbers are written in parenthesis.

| Dataset | Threshold | Nodes | Edges | Average degree | Clustering |
|---|---|---|---|---|---|
| No al Tarifazo | 3 | 15.67 % (123) | 8.32 % (753) | 12.24 | 0.80 |
| 9n | 5 | 13.08 % (167) | 9.02 % (1541) | 18.46 | 0.77 |
| Charlie Hebdo | 3 | 6.82 % (119) | 3.0 % (271) | 4.55 | 0.51 |

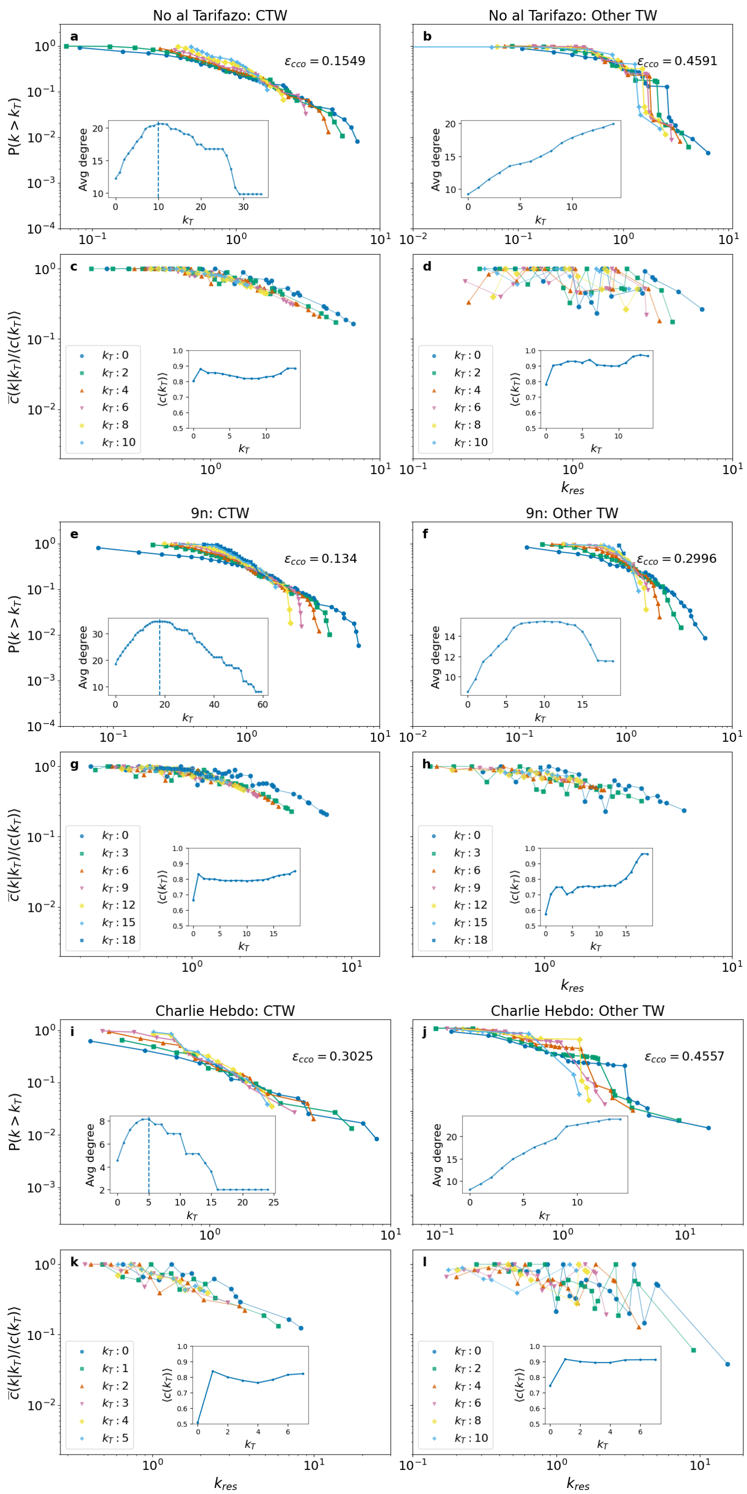


FIG. SI.3: **Complementary cumulative distribution function and average clustering coefficient for critical and non-critical temporal windows.** Each row of panels compares the CTW (left column) against a representative non-critical temporal window (right column) for the same dataset. The top row of each pair presents the CCDF of the normalised node degree for various degree thresholds $k_t$, with the inset showing the evolution of the average degree as a function of $k_t$. The bottom row displays the average clustering coefficient as a function of $k_{res}$. The $\varepsilon^2_{cco}$ values reported in each panel quantify the degree of curve collapse: low values indicate self-similar behaviour, whilst high values reflect the absence of collapse. In all three datasets, the CTW yields substantially lower $\varepsilon^2_{cco}$ values and clearer collapse of both the CCDF and clustering spectrum curves compared to non-critical windows, confirming that self-similarity is a distinctive structural signature of peak mobilisation. From top to bottom: No al Tarifazo, 9N, and Charlie Hebdo datasets.